Multi-Step Deformable Object Manipulation via Adaptive State Abstraction for Model-Based Control

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***Abstract*— We propose a hybrid controller that reasons over different levels of state abstraction to solve long-horizon manipulation tasks. Performance of these tasks crucially depends upon the information conveyed in low-dimensional state representations, whether they are handcrafted or learnt. Unsolved perception challenges involve identifying high-utility features for myriad tasks, and grounding these features in sensor observations. Existing reinforcement learning (RL) approaches learn and fix task-specific representations, which places limitations on the agents, which will inevitably encounter task variations. Motivated by recent developments in representation learning and hierarchical RL, in this work we address multi-step deformable object manipulation using adaptive representations in a model-based RL framework. Results from our experiments demonstrate notable performance improvement over fixed-representation baselines in cloth reconfiguration tasks.**

I. Introduction

Deep learning in the robotics domain has enabled decision-making algorithms to operate directly over high-dimensional input spaces (e.g., images), voiding the need to manually design state representations [14, 24]. Researchers working with increasingly complex tasks have pivoted toward low-dimensional representations for use in reinforcement learning (RL), planning, and control frameworks [20, 21, 41]. For unstructured environments, however, determining what information should be summarised in the low-dimensional states is a non-trivial problem. Solutions that learn and fix representations are limited when visual or physical aspects of the task vary [7].

This paper addresses multi-step deformable object manipulation tasks where extensive deformation presents modelling challenges. Though task execution need not rely upon explicit models, it is of interest to use model-based methods for their efficiency and robustness to data distribution shift. However, the high-dimensional state space and nonlinear dynamics of the objects introduces challenges in controller optimisation under large object deformation [9, 15]. Some of these challenges can be addressed by demonstration tracking, a problem that we frame as sequential goal-conditioned object reconfiguration.

Our approach to object reconfiguration is two-pronged: 1) solve a *task-conditioned* representation selection problem at a high-level, and 2) optimise a low-level control trajectory. We propose a model-based hierarchical controller that modulates the granularity of the state information used to solve a sequence of goal-conditioned steps. The controller's notion of state takes inspiration from recent advances in self-supervised representation learning in order to distill different abstractions of task-relevant state information [10, 14]. With an adaptive state, the controller can attend to relevant features across different sub-tasks – e.g., spreading versus unfolding a piece of fabric.

The research contributions of this work include:

- An analysis of the trade-offs between employing different dimensional latent vector representations for solving short-horizon and multi-step tasks.
- An implementation of a proof-of-concept adaptive control framework that dynamically filters state information, tailoring trajectory tracking solutions to local parts of the state space.
- An evaluation that demonstrates the advantage of executing long-horizon tasks using adaptive representations as opposed to fixed representations.
- A physical system implementation that tests our proof-of-concept in a real-world setting[1].

II. Background

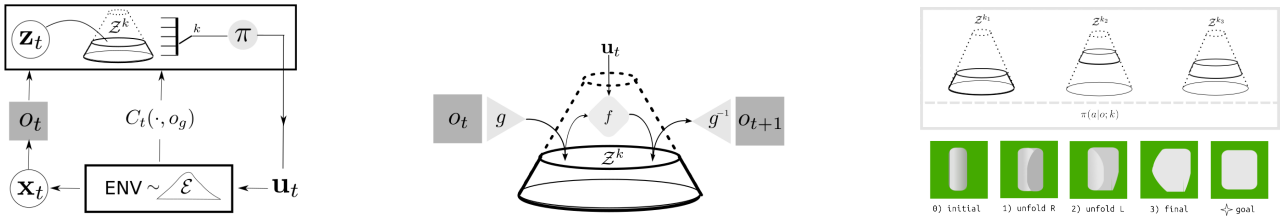
Prior deformable object manipulation systems have used hand-crafted features to define states for traditional planning and control frameworks. Early work proposed graphical and topological state descriptions [17, 26, 36, 40], with mass-spring and mesh models later developed for use in quasi-static [3, 13, 28, 36, 37, 42] and dynamic object shaping [4, 26, 39]. However, high aleatoric uncertainty and cumbersome parameter tuning limit the scalability of these representations; we instead seek representations that can adapt to unanticipated physical and visual variations of the object.

Data-driven methods have previously been successfully applied to learn task-relevant features for low-level deformable object manipulation. Contrastive learning, for example, provided an avenue for self-supervised learning of object descriptors [18, 31] and low-dimensional dynamics models [14, 21, 23, 41]. Other approaches have relied upon supervised learning for key point detection [2, 35] in dense object configurations. A limited subset of this literature extends these techniques to tasks beyond long-horizon rope manipulation. Visual foresight models have been applied to cloth-folding tasks [8, 29], but risk compounding errors by optimising in the pixel space.

Typically, a hierarchical approach is required in order to reason about long-horizon tasks. Many approaches have shown impressive results on complex object reconfiguration

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Hybrid controller | Dynamics model | Adaptive state for MBRL

Fig. 1: Our proposed adaptive controller π and dynamics model is designed to address variations in the environment due to extensive object deformation. Control modes k define different levels of state abstraction.

using learning from demonstration methods to guide low-level task execution [1, 22, 27, 28, 38]. Unfortunately, a sufficient distribution gap between the training and deployment contexts can degrade performance, making these methods potentially poors choice for long-horizon settings. Our system, on the other hand, is designed with a large degree of task variation in mind.

Ideas around model and skill selection have been proposed to solve long-horizon tasks (e.g., cloth draping [19]) without human guidance. One notable innovation, DiffSkill [16], used hierarchical RL to complete food preparation tasks by optimising over model-based policies, or skills; for example, manipulating tools to roll out and cut dough. Contrary to the task domains explored in that work, however, the higher degree of precision in cloth manipulation necessitates solving a non-trivial state estimation problem. Thus, we focus on leveraging abstractions in the state space, as seen in recent trial-and-error schemes [10, 11].

III. Multi-Step Object Reconfiguration

In this work, we present an alternative approach to solving multi-step deformable object manipulation tasks via a control hierarchy that leverages time-varying state representations. We formulate these multi-step tasks as trajectory tracking problems.

A. Preliminaries

Sequential decision making problem is commonly modelled as Markov decision process (MDP). Analogously, the control theoretic view frames the problem using a controller-plant model with the following notation: $\mathbf{x} \in \mathcal{X}$ states , $\mathbf{u} \in \mathcal{U}$ control inputs, $c : \mathcal{X} \to \mathbb{R}$ cost, and $f : \mathcal{X} \times \mathcal{U} \to \mathcal{X}$ dynamics, with $\mathcal{X} \subset \mathbb{R}^n$ and $\mathcal{U} \subset \mathbb{R}^m$. Our problem of interest assumes unknown dynamics, resulting in an RL problem [32].

B. Model predictive control

Our approach embraces the model predictive control (MPC) paradigm, in which dynamics estimation serves to restrict the search space over the decision variables [25]. More concretely, we ground tasks in the sequential optimal control problem (OCP) formulation.

Let $\tau = \{\mathbf{u}_0, \ldots, \mathbf{u}_{N-1}\}$ represent a trajectory of discrete-time control inputs, f_θ denote a deterministic approximation of the dynamics (e.g., by a neural model with weights θ), and $\ell, V_{\text{final}} : \mathcal{X} \to \mathbb{R}$ be instantaneous and terminal cost functions. A controller $\pi(\mathbf{u}_t|\mathbf{x}_t)$ seeks to achieve the following objective:

$$\min_{\tau} \sum_{t=0}^{N-1} \ell(\Phi(t; \mathbf{x}_0, \tau)) + V_{\text{final}}(\Phi(N; \mathbf{x}_0, \tau)) \tag{1}$$

subject to

$$\mathbf{x}[0] = \mathbf{x}_0$$
$$(\Phi(t; \mathbf{x}_0, \tau), \tau_t) \in \mathbb{Z}_{feasible}, \forall t$$
$$\Phi(N; \mathbf{x}_0, \tau) \in \mathcal{X}_{final}$$

provided an initial state $\mathbf{x}_0$, time horizon $N < \infty$, a set of admissible trajectories $\mathbb{Z}_{feasible}$, and a set of terminal states $\mathcal{X}_{final}$. The function $\Phi : \mathbb{R} \to \mathcal{X}$ represents a forward simulation function (running in the interval $j = [0, t]$) that enforces dynamical constraints. Latent variable models can enable optimisation over abstractions of high-dimensional states and nonlinear dynamics using established control frameworks, as we describe in the following sections.

C. Problem formulation

We extend the MPC formulation to address multi-step deformable object reconfiguration tasks in a tabletop setting. To mimic real-world task execution, we assume that π does not directly access states, but visual observations given by a state-emission distribution $p(o|s) \in \mathcal{O}$. The policy π must then solve a partially observable Markov decision process in which it aims to reproduce a sequence of image observations produced by an oracle policy. The oracle $\pi^*(\mathbf{u}_t|o_t)$ provides an M-length sequence of sub-goals $S_g^M = \{o_0, o_g^1, \ldots, o_g^M\}$ that transform the object from an initial configuration o_0 to a target configuration $o_g^* = o_g^M$. The sub-goals do not explicitly contain any information about the oracle's actions.

In this problem variant, a controller $\pi(\mathbf{u}_t|o_t)$ optimises a trajectory τ of non-planar end-effector poses, $\mathbf{u}_t \in \text{SE}(3)$ or $\mathbf{u}_t \in \mathbb{R}^6$ in pixel space, according to:

$$\min_{\tau} \sum_{i=1}^{M} \sum_{t=0}^{N-1} \ell(\Phi(t), o_g^i) + V_{\text{final}}(\Phi(N), o_g^i) \tag{2}$$

subject to

$$o_{t=0}^1 = o_0 \tag{3}$$
$$(\Phi(i * t), \tau_{(i*t)}) \in \mathbb{Z}_{feasible}, \forall i, \forall t \tag{4}$$
$$\Phi(i * N) \in S_g^M, \forall i \tag{5}$$

Algorithm 1 Hybrid Control

procedure RUN(S_g^M, $\mathcal{K}$, N) ▷ note: t is time step
  $\tau = \emptyset, k = \min(\mathcal{K})$ ▷ init. trajectory, mode
  $h_\theta^k = (g^k, f^k)$ ▷ and dynamics model
  for $i = [1, \ldots, M]$ **do**
    $z_g^i \leftarrow g^k(o_g^i), t \leftarrow 0$ ▷ encode goal obs
    for $t = [1, \ldots, T]$ **do** ▷ execute step with
      $z_t \leftarrow g^k(o_t)$ ▷ variable modes
      $k =$ SelectMode($G(z_t, z_g^i), \mathcal{K}$)
      $\tau \leftarrow \tau \cup$ RunMPC(o_t, o_g^i, N, k)
  return τ

where $\Phi(\cdot) = \Phi(\cdot; f_\theta, o_0^1, \tau)$ is shorthand for the forward simulation function. The controller is evaluated at each step according to the goal-conditioned costs $\ell, V_{\text{final}} : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$.

Going forward, prime notation is used as short-hand for the subsequent time step; i.e., $o' = o_{t+1}$.

IV. A Hybrid Control Approach

Lack of sparsity in the optimal control search space motivates our treatment of the multi-step object reconfiguration problem as trajectory tracking. Notably, replicating the reference trajectory S_g^M is complicated by the high dimensionality of the underlying state space. Extensive object deformations also incur visual and physical variations during execution. Optimising a control trajectory over hierarchical state abstractions can address these problems. We design a hybrid controller that invokes different control modes, implemented as low-level policies defined over different levels of abstractions, to solve long-horizon tasks.

A. Controller design

The hybrid controller relies on an adaptive observation model and a latent dynamics model to optimise the control inputs. Let $\mathcal{K} = \{2^3, 2^4, \ldots, 2^{12}\} \subset \mathbb{N}$ be a set of feature sizes and $k \in \mathcal{K}$ specify the dimensionality of an embedding space $\mathcal{Z}^k \in \mathbb{R}^k$. A joint observation and dynamics model $h_\theta : \mathcal{O} \times \mathcal{U} \to \mathcal{Z}^k$ gives an approximation of state-transitions w.r.t. which control inputs are planned. Emulating an adaptive representation, we define a hierarchical control scheme (Alg 1) using a set of models $\{h_\theta^k | k \in \mathcal{K}\}$. A high-level controller optimises over control modes that determine which low-level policy $\pi_k(\cdot|o; h_\theta^k)$ is active at each time step. It yields a control mode trajectory $\lambda = \{k_t^i | i \in [1, M], t \in [0, T-1]\}$ according to Eqn 2, but with horizon $N = 1$ and the modified objective:

$$\min_{\lambda, \tau} \sum_{i=1}^{M} \sum_{t=0}^{T-1} \ell(\Phi(t), o_g^i) + V_{\text{final}}(\Phi(T), o_g^i)$$

where $\Phi(t) = \Phi(t; f^{(\cdot)}, o_{t-1}, \tau)$ and ℓ, V_{final} are a function of embeddings $z \in \mathcal{Z}^{(\cdot)}$. The RunMPC routine executes a low-level policy using model $h_\theta^k = (g_\psi, f_\phi)$ to encode observations into a k-dimensional latent space and to forecast subsequent latent trajectories. Note that $\mathbf{u}$ is defined in pixel space.

The mode selection scheme uses state-dependent criteria. As a simple example, consider a criterion based on either the state trajectory's convergence (equivalently, its velocity) $G_{\epsilon\text{-conv}}(z_t, z_{t+1})$ or its proximity to a sub-goal $G_{\epsilon\text{-prox}}(z_t, z_g)$. A binary multiplexer may use criteria based on $G(z, z')$ to switch between high-k and low-k modes:

$$G(z, z') = \begin{cases} 1, & \text{if } \|z - z'\|_2^2 \leq \epsilon \\ 0, & \text{otherwise.} \end{cases}$$

Given a more elaborate selection procedure, the hybrid controller can carry out a coarse-to-fine strategy to quickly drive states to the goal configuration, increasing k to refine the latent structure for precise alignment and stabilisation around the goal.

Upon selecting mode k, the hybrid controller engages an MPC controller defined over a k-dimensional state space. Following methods from Yan *et al*. (2020), we use Cross-Entropy Method (CEM) to optimise the low-level control inputs. This randomised algorithm yields control inputs $\mathbf{u}_t^*$ that minimise the future cost $\sum_{i=0}^{n-1} \|f(z_i, \mathbf{u}_i) - z^*\|_2^2$ while avoiding local optima. As described further in Section V-C, the baseline controllers in our experiments were implemented using fixed $k = 8$ and horizon $N = 1$.

B. Observation and action space

The observation space consists of images $o \in \mathbb{R}^{W \times H \times c}$ of width W, height H and (channel) depth c. The observations are captured from a birds-eye view of the scene. As part of pre-processing, the pixels are normalised and transformed to the range $[-1, 1]$. These transformations ensure that model inputs are zero-centered and comparable, desirable properties for model training.

The control input space consists of normalised pixel indices $[-1, 1]^6$ that determine pick location (u, v, d) and displacement vector in a depth image. We use a segmented depth image to restrict the action space to points on the object, making training more efficient. Section describes how to map pixel space control inputs to SE(3).

C. Dynamics estimation

Convergence of MPC algorithms hinges on low dynamics model error, a condition that is difficult to satisfy given deformable objects. The complexity of these systems, i.e., under-actuation and high-dimensionality, makes learning representations with supervised methods impractical. Contrastive learning methods, on the other hand, derive a learning signal from dissimilarities between clusters in the training data. This learning mechanism is appealing for finding useful state representations that are unintuitive, or otherwise difficult to specify.

We use contrastive learning to train an ensemble of models $\{h_\theta^k\}$ to make forward predictions in the latent space that are consistent with observed dynamics $o' = f(o, \mathbf{u})$ [41]. A dynamics data set $\mathcal{D}$ is generated by collecting rollouts from random policies in DeepMind control suite (DM control suite) environments [33, 34]. State-action trajectories are processed into samples $\mathrm{d} = (o, o^+, o^-, \mathbf{u})$ consisting of

an observation o, action $\mathbf{u}$, positive example $o^+ = o'$ and negative examples $o^- = \{o_i\}_{i=[1:B]}$, where B is the batch size minus one. The negative samples are a set of observations sourced from randomised batches drawn from $\mathcal{D}$.

The training objective optimises a latent space in such a way that preserves mutual information between latent state-trasitions and corresponding environment dynamics [23]. Recall that $\mathrm{d} = (o, o^+, o^-, \mathbf{a})$ is a dynamics data point. Models $\{h_\theta^k\}$ are independently trained according to:

$$\mathcal{L}_{\text{contrastive}} := -\mathbb{E}_{\mathrm{d}\sim\mathcal{D}}\left[\log \frac{\mu(z^+, z')}{\sum_{i=1}^{B} \mu(z^+, z_i^-)}\right], \qquad (6)$$

with prediction $z' = f_\phi(o_t, \mathbf{a}_t)$, positive sample $z^+ = g_\psi(o^+)$ and negative samples $z^- = \{z_i = g_\psi(o_i^-)\}_{i=[1:B]}$. A carefully chosen similarity measure and forward prediction task can guide learning of a latent structure that captures the observed dynamics. As in Yan *et al*. (2020), we use the squared exponential similarity measure

$$\mu(z_1, z_2) = \exp(\|z_1 - z_2\|_2^2).$$

Alongside the forward model, we optimise an observation model ${g_\phi}^{-1}$ using a standard mean-squared error reconstruction loss (Section V-D).

V. Experiments

Our experiments begin by investigating the trade-offs between model variants; i.e., w.r.t. choice of similarity metrics, dimensionality of the embedding space, and other training hyper-parameters. Initial findings motivate a key design choice for the adaptive control prototype: to use the dimensionality of the latent space to affect the granuality of information captured in representations. In further experiments, the hybrid control design is evaluated on a set of multi-step tasks against fixed-representation baselines.

A. Experiment setup

The experiment setup consists of a table-top scene with a solid background and a set of objects for different reconfiguration tasks inspired by past work on human-guided RL systems [8, 22, 29, 30]. We evaluate controller performance in a short-horizon domain, focussing on model evaluation, and a tracking domain.

B. Short-horizon task suite

Each task in the short-horizon task suite involves shaping a single target object into a target configuration:

Random Rope A rope must be arranged into an arbitrary shape. It's initial configuration is randomised.

Smooth Cloth From a random initial state, a cloth must be shaped into a flat square at the center of the frame.

Random Cloth A cloth must be shaped into a random target configurations given random initialisation.

These short-range tasks provide a test bed for analysing model predictive performance and controller performance. The rope shaping experiment captures the simplest case in which we can easily interpret model behaviour. We allot controllers 20 time steps to complete rope shaping tasks, and 40 time steps for cloth tasks. The domain drift between model training and evaluation is assumed to be minimal, though addressable by domain randomisation.

C. Tracking task suite

This collection of multi-step tasks tests the adaptive controller's capability to reason about extensive object deformation:

Cloth Center Unfold A three-way center fold must be undone into a flat configuration.

Cloth Square Unfold A flat configuration is achieved by unfolding along the horizontal axis, then the vertical axis.

Cloth Fold A sequence of folds along the vertical axis is required to shape a smooth cloth into a narrow rectangle.

These multi-step tasks capture challenging object configurations that illustrate the need to trade-off state abstractions. Some configurations require a coordinated sequence of actions to align folds, or may benefit from locating corners and edges in task phases with high self-occlusion. Accomplishing the tasks efficiently motivates the use of state abstractions that focus on high-utility features, as determined by the task phase. We construct demonstrations sequences of length $M = 4$ for all tasks and we allot 40 time steps to reach each sub-goal. The initial state is fixed for each task.

Sub-goal sequences S_g^M for each tracking task are sampled from an oracle policy. An object is initialised in the goal configuration, and the policy reconfigures its shape using a task-specific state machine, with key point-based states. As an example, a Cloth Folding oracle alternates between picking the top left corner and the bottom left corner key points, applying a large horizontal force to each corner to complete a single fold. We construct a demonstration by subsampling and reversing the oracle's state trajectory.

D. Model & controller evaluation

1) Overview of metrics: We evaluate models according to forward prediction error and policy performance as measured by:

Mean-squared error, a standard pixel-wise similarity measure between observations. $\text{MSE}(o, o) := \sum_{u,v}(o_{u,v} - o'_{u,v})^2$

L2-norm distance as a measure of disparity between latent states, written as: $\text{L2}(\mathbf{x}_1, \mathbf{x}_2) := \|\mathbf{x}_1 - \mathbf{x}_2\|_2^2$

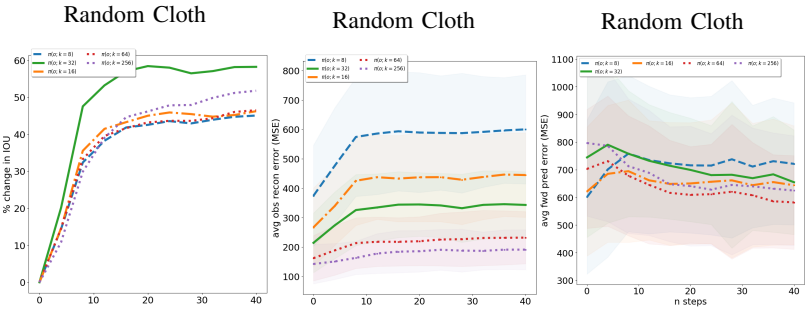
Jaccard distance (Intersection-over-Union, IoU) as similarity between object configurations, assuming that foreground objects are distinct from the background. Letting $m(\cdot)$ be a function that returns a mask-segmentation of the object in o, $\text{IoU}(o, o') := \sum_{u,v}(m(o) \cap m(o'))_{u,v} / \sum_{u,v}(m(o) \cup m(o'))_{u,v}$

Percent change in goal distance (w.r.t. the above metrics), written as: $\Delta(\mathrm{m}; \mathrm{m}_0) := 100 \cdot \frac{m - m_0}{\max(\varepsilon, m_0)}$, with $\varepsilon > 0$, m_0 as the initial distance metric and m the current metric.

2) Model evaluation: The contrastive training loss (Eqn 6) and reconstruction error are early indicators of model predictive performance. The contrastive loss represents the degree to which latent states transitions correspond to the observed dynamics, e.g., by maximising mutual information [23]. On

	C_{final} $(\pm\sigma)$ ↑		
Hyper-Parameters	Smooth Cloth	Random Rope	Random Cloth
Yan *et al.* (2020)	n/a → 59(7)	n/a	n/a
k=8, c=4	53 → 61.4(13)	1.6 → 5.9(3.8)	39 → 57.8(11)
k=16, c=4	53 → 63.1(13)	1.2 → 6 (3.7)	41 → 60.7(10)
k=32, c=4	53 → 63.6(12)	1.9 → 6.7(5.5)	38 → 60.6 (9)
k=64, c=4	53 → 64.2 (12)	1.7 → 5.4(4.7)	43 → 63.8(9)
k=256, c=4	52 → 60.3(13)	-	39 → 59.7(11)
k=16, c=1	74 → 74.9(7)	-	-

TABLE I: Final raw goal coverage (x100) from initial.



(a) Goal Coverage ↑ (b) Reconstruction Error ↓ (c) Prediction Error ↓

Fig. 2: Performance on short-horizon tasks: a) average goal coverage (%ΔIoU) over time steps for policies with fixed $k \in \{8, 16, 32, 64, 256\}$, b) reconstruction error (MSE) and c) forward prediction error (MSE) for corresponding models.

the other hand, the reconstruction error, $\text{MSE}(o_t, {g_\phi}^{-1}(\mathbf{z}_t))$, provides a proxy for information lost during compression by g_ψ.

In addition to the prediction task at training time, our models are evaluated based on their utility in solving trajectory optimisation. All task performance metrics are computed using expected goal coverage $C_t = \text{E}_{(o_t, \mathbf{a}_t)\sim\pi}[\text{IoU}(o_t, o_g)]$. We also use expected goal distance $L_t = \text{E}_{(z_t, \mathbf{a}_t)\sim\pi}[\text{L2}(\mathbf{z}_t, \mathbf{z}_g)]$ to understand the effect of model quality on latent planned trajectories, but we compare performance using metrics in pixel space.

3) Baseline policies: The baseline controllers for the short-horizon suite use models with RGB input ($c = 3$) and 8-dimensional latent states [41]. In the tracking suite, we use policies that perform well in the short-horizon setting as baselines. These baselines, denoted by $\{\pi_{\text{fixed}}(o; k = 8), \pi_{\text{fixed}}(o; k = 32)\}$, use fixed representations generated by "high-level" and "low-level" models trained with RGB-D inputs ($c = 4$).

VI. Evaluation Results

1) Model Performance: Our initial experiments tested models trained with alternative sensor inputs, (fixed) latent space dimensionality, similarity metric μ and training duration. The goal was to discover variations that could provide a useful mechanism for changing the hybrid controller's state representation. Due to page limit restrictions, we focus attention on variations that increased task performance; in particular, inclusion of depth input and modulation of the latent space dimensionality.

We found that the optimal latent dimensionality varied from task to task. For the Random Rope shaping task, increasing the dimensionality to $k \in \{16, 32\}$ yielded up to +30% ΔIoU. Instances in which policies were required to with shaping the rope into a flat configuration did not benefit from increasing k. Interestingly, cloth shaping policies saw higher performance with $k \in \{16, 32, 64, 256\}$. The final goal coverage (Table I) in Smooth Cloth increased up to 64% for models with increased dimensionality, nearly a 10% rise over baseline in Yan *et al.* (2020). Figure 1b shows an example of the effect of increasing k on reconstruction error.

2) Controller Performance: Experiments in the trajectory tracking suite indicated that the coarse-to-fine strategy carried out by hybrid controllers typically outperformed the baselines. Hybrid controllers π_{conv} and π_{prox} used models trained with $k_1 = 8$ and $k_2 \in \{8, 32, 64\}$ and mode selection criterion with $\epsilon \in \{1, 1.5, 4\}$. The initial configuration was sufficiently far from the goal in all tracking suite tasks (Figure 4).

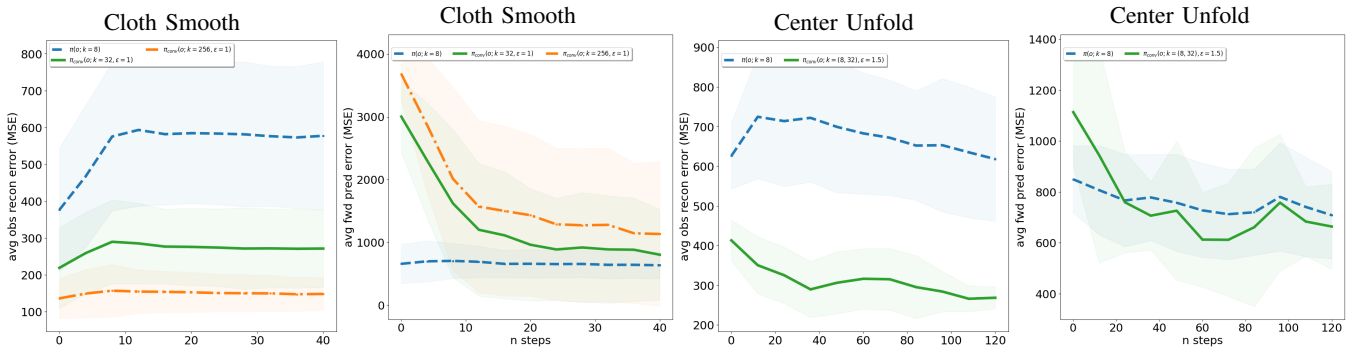
Goal coverage metrics summarised in Table II and Figure 4 indicated that, over 500 trials, both baseline controllers were sub-optimal compared to the minimal k hybrid controllers. The coarse-to-fine strategy lead to a 14% increase in ΔIoU over $\pi_{\text{fixed}}(o; k = 32)$ to achieve 47.5% goal coverage in the Cloth Square Unfold task (Table II). Figure 4 illustrates tracking results for $\pi_{\text{fixed}}(o; k = 8)$ (center) and the hybrid controller $\pi_{\text{conv}}(o; h^{\{8,32\}}, \epsilon = 4)$ (right) on the Cloth Center Unfold task. The baseline failed to maintain the shape of the cloth. The hybrid controller, on the other hand, managed to maintain the cloth's orientation, and in the final step successfully spreads the cloth from a highly self-occluded configuration.

TABLE II: Final raw goal coverage (x100) for baseline and hybrid controllers. Hybrid controllers used either $G_{4\text{-prox}}$, $G_{1\text{-conv}}$ or $G_{1.5\text{-conv}}$ as selection criteria.

# trials=500	$C_{\text{final}}(\pm\sigma)$ ↑	
Controller Type	Cloth Center Unfold	Cloth Square Unfold
$\pi_{\text{conv}}(\mathbf{z}; h^{\{8,32\}})$	58.1 (6.2)	47.5 (7.8)
$\pi(\mathbf{z}; h^{8})$	52.1 (8.3)	41.6 (8.1)
$\pi(\mathbf{z}; h^{32})$	51.7 (8.8)	45.1 (6.4)
$\pi_{\text{prox}}(\mathbf{z}; h^{\{8,32\}})$	54.0 (8.5)	45.7 (7.0)
$\pi_{\text{prox}}(\mathbf{z}; h^{\{32,64\}})$	50.6 (4.0)	42.4 (7.3)

VII. Discussion

This collection of experiments revealed the surprising utility of modulating latent space dimensionality to tune the state representation for a particular task. This insight allowed us to design a hierarchical controller that reasoned over multiple levels of state abstractions to accomplish long-horizon tasks. The proposed controller achieved higher coverage scores



(a) Reconstruction error ↓ (b) Forward prediction error ↓ (c) Reconstruction error ↓ (d) Forward prediction error ↓

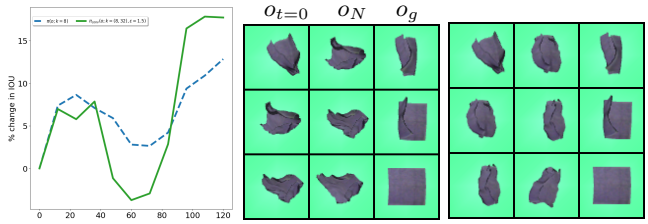
Fig. 3: Comparison of model performance on short-horizon (a,b) and multi-step tasks (c,d). The baseline π_{fixed} (blue) was configured with $k = 8$. The hybrid model π_{conv} (green) alternated between modes $k_1 = 8$ and $k_2 = 32$ using the convergence criterion with $\epsilon = 1.5$. MSE is reported for observation reconstruction and forward prediction.

than fixed-representation baselines across 5 out of 6 short-horizon and multi-step tasks.

1) Predictive performance: The short-horizon task suite revealed that increasing k aided models in forward prediction and observation reconstruction, especially in cloth shaping tasks. The reconstruction error nearly halved with 4x increase in k in the Random Cloth task (Figure 1b). We suspect that models trained with higher k learnt features that conveyed more fine-grained state information. Similar trends were observed in the tracking setting (Figure 3).

2) Task performance: Counter-intuitively, increasing k, seemingly refining latent features, did not guarantee improved performance. Figures 2b-c suggest that high k should be preferred for modelling the cloth state, yet controllers under these models under-performed those with $k = 32$. This observation implies that highly expressive representations can be detrimental to task performance where a small collection of features matter [14]. In the smoothing task, controllers using high k representation may unproductively choose small actions to manipulate small folds in a bundle of cloth.

The advantage of using varied k over minimalist representations became clear in the multi-step setting, where the collection of useful features is not fixed (Figure 4).



(a) Goal coverage (b) $\pi(\mathbf{z}; h^8)$ (c) $\pi(\mathbf{z}; h^{\{8,32\}})$

Fig. 4: Average goal coverage (%ΔIoU) over time for a baseline (b) and hybrid controller (c) on tracking the cloth center unfold demo. Each row corresponds to one step in the task.

The Cloth Unfold tasks required controllers to stabilise trajectories about each sub-goal to achieve the final goal in time. Qualitatively, the hybrid controllers were able to trade-off coarse and fine abstractions to achieve the goal quickly and more precisely.

3) Selection strategies: Out of the simple selection schemes introduced in Section IV-A, we found the convergence criterion to be most useful in the cloth domain. Table II showcases the advantage of tracking using state trajectory convergence as a signal for switching between levels of abstraction. Possibly, such a trade-off allows the hybrid controller to escape local minima found while optimising over coarse features. This insight could frame a more sophisticated selection scheme in future work.

4) Limitations: Our proposed hybrid controller establishes a clear benefit of doing long-horizon tasks with adaptive state abstractions. More pronounced performance gains would be achieved given a less myopic RL algorithm; e.g., model-based actor-critic agent [5, 6]. Further, a cleverly designed hierarchical forward model, perhaps as an extension of Kusupati *et al.* (2022) rather than an ensemble, may drastically reduce the cost of training [12]. We leave this problem for future pursuit.

VIII. Conclusion

This paper introduced a model-based RL approach to solving multi-step manipulation tasks with high dimensional systems. The controller leveraged an adaptive representation to reason over large deformations of a deformable object. Initial model evaluation uncovered the surprising trade-offs of increasing the latent space dimensionality; namely, lacking performance due to overly expressive representations. Further experiments showed that a simple hybrid control scheme could successfully trade-off the advantages of coarse and fine representations to outperform controllers with fixed representations. Future directions in this problem space should investigate unified hierarchical dynamics models and enhanced mode selection algorithms.

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